

Illustration by TANMOY CHAKRABORTY

PADDY SHOOTS, USED AS FODDER, CONTAIN ARSENIC, THUS PUTTING AT RISK HUMANS WHO CONSUME THE MEAT/ MILK OF COWS AND GOATS

In contrast to arsenic poisoning, humans are squarely responsible for the chemical contamination of fish, and it comes from formalin, or formaldehyde dissolved in water, otherwise used as a disinfectant in industries as well as to embalm and preserve corpses.

West Bengal's ministry of fisheries has asked the Andhra Pradesh government to watch out for the use of formalin for preservation of Rohu and Katla fish being caught from Nagarjun Sagar, Bhimavaram, Srisailam and Koleru lake reservoirs. Andhra, the largest fish-producing state in India, exports the majority of its produce to the Bengal market. Fish transported to West Bengal needs refrigeration twice on its way. Refrigeration of one batch of 40-50 kg fish requires 10 kg ice. The use of formalin reduces that requirement by half. Formalin is injected in the abdominal region of fish, contaminating them thoroughly. Their appearance, however, is unaffected. Officials in the state fisheries department admit that some unscrupulous fish traders often use formalin to preserve fish, but claim that surprise raids have reduced the habit.

Again, many fish farmers use leather extracts and leather waste as fish feed. Leather waste is laden with chromium, lead and zinc, which are heavy metals and toxic chemicals. "These heavy metals get accumulated in the muscle, as they are not fat-soluble, unlike most pesticides. This is a bigger threat," says an expert from the ministry of fisheries. Though there has been increased surveillance to curb the practice, experts are sceptical if it can be stopped completely. "I know many fish ponds in North 24 Parganas where tangra (*mystus cavasius*) and other fish are made to nibble at leather waste to help them grow in size," says a state government official.

The threat of contamination of fish also comes from mercury, another toxic chemical, which occurs in abundance in industrial and medical effluents dumped in rivers and water bodies. Mercury gets transformed into the highly toxic methyl mercury through microbial action and settles on water planktons, which are food for small fish. Bigger fish consume smaller fish, and are in turn eaten by humans, thus entering the food chain. The human nervous, digestive and immune systems can be severely damaged by the intake of mercury through food. Though mercury contamination of fish was brought to the state government's attention by NGOs a decade earlier, some experts say the situation is not grave. Others say the ingress of sea water during high tide significantly dilutes any such contamination in the rivers in south Bengal.

Is this concatenation of alarming facts food for thought for the average Bengali content with her *maachh-bhaat*? Hopefully, she will stop to think before digging into her comfort food. ■